

Network Topologies

Network Topologies

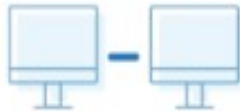
- Refers to how various **nodes, devices**, and connections on your network are **physically** or **logically** arranged in relation to each other.
 - The **physical topology** of a network refers to the physical layout of the devices, but mostly the cabling and cabling layout.
 - The primary physical topology categories are **bus, ring, star**, and **mesh**.
 - You must match the physical topology to the type of cabling (twisted pair, coaxial, fiber and so on) that will be installed.
 - The **logical topology** defines the logical path on which the signal will travel from on the physical topology.
 - that is the way in which data accesses the network media and transmits packets across it.
 - is bound to **network protocols** and describe how data is moved across the network

Why we need a topology ?

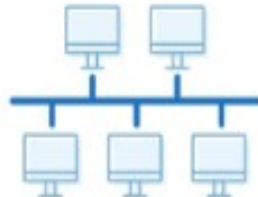
- A network's topology affects its capabilities
- The choice of one topology over another will have an impact on the
 - Types of equipment that the network needs
 - Growth of the network – scalability
 - Way the network managed

Network Topology Types

1 Point to point



2 Bus



3 Ring



4 Star



5 Tree

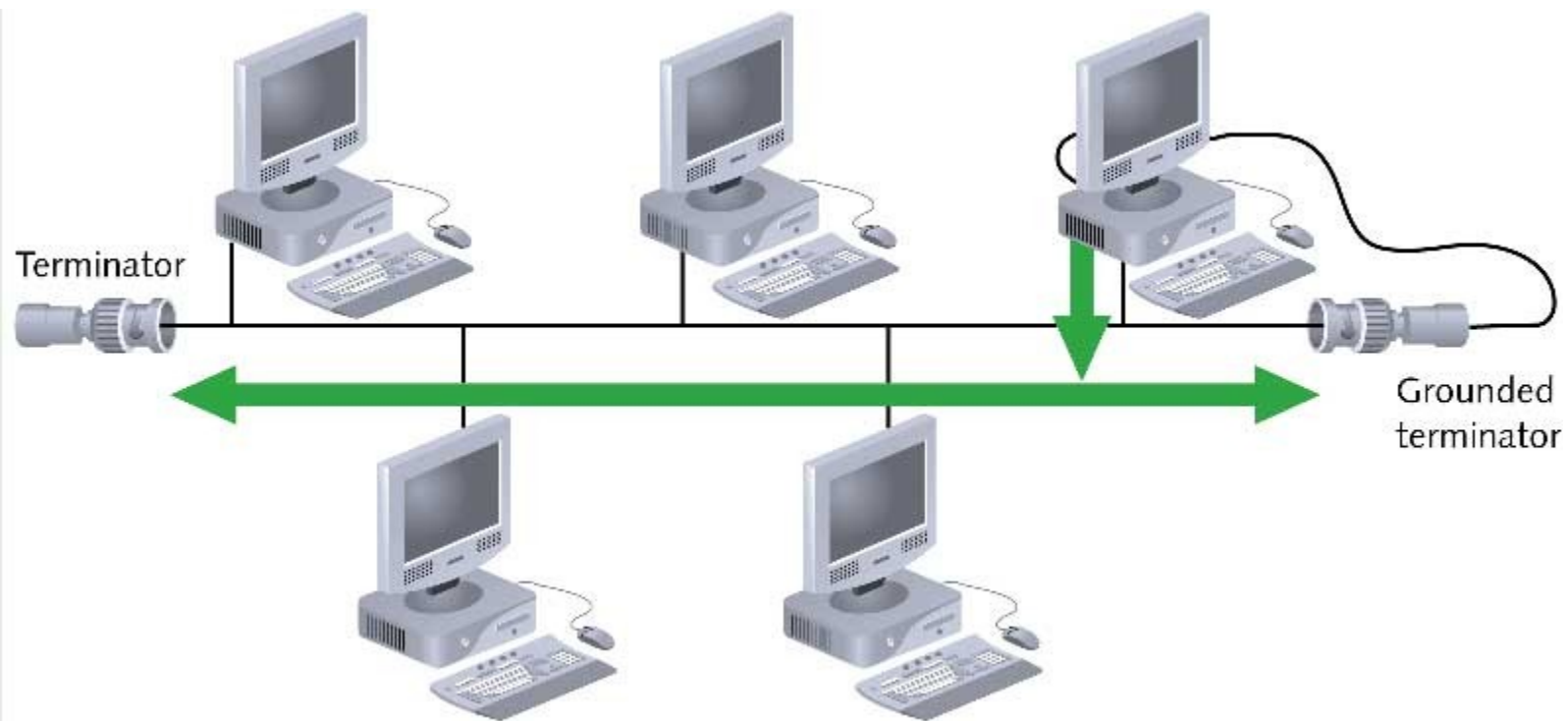


6 Mesh



7 Hybrid





Bus Topology

- A **Bus topology** consists of a single cable—called a **backbone**— connecting all nodes on a network without intervening connectivity devices
- Linear bus
- one long cable acts as a backbone to link all the devices in the network
- All the device receive the signal.
- It describes a multipoint connections
- Cable must end with a terminator that absorbs the signal when it reaches the end of the line wire.

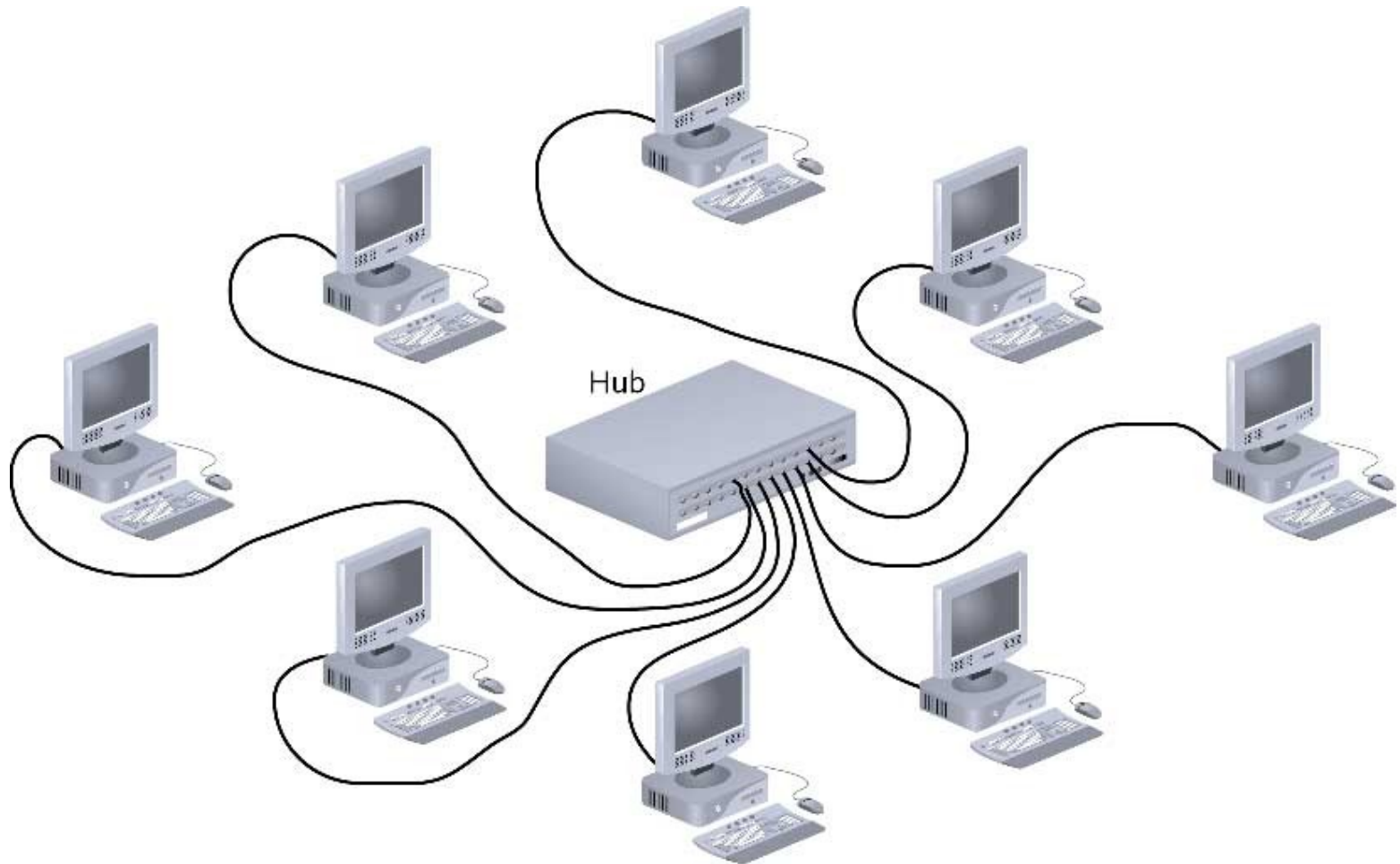
- **Advantages**

- ease of installation; less cabling
- less cabling ; If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Bus topologies are a good, cost-effective choice for **smaller networks** because the layout is simple, allowing all devices to be connected via a single **coaxial** or **RJ45** cable.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- Bus topology is familiar technology as installation and troubleshooting techniques are well known.
- If needed, more nodes can be easily added to the network by joining additional cables.

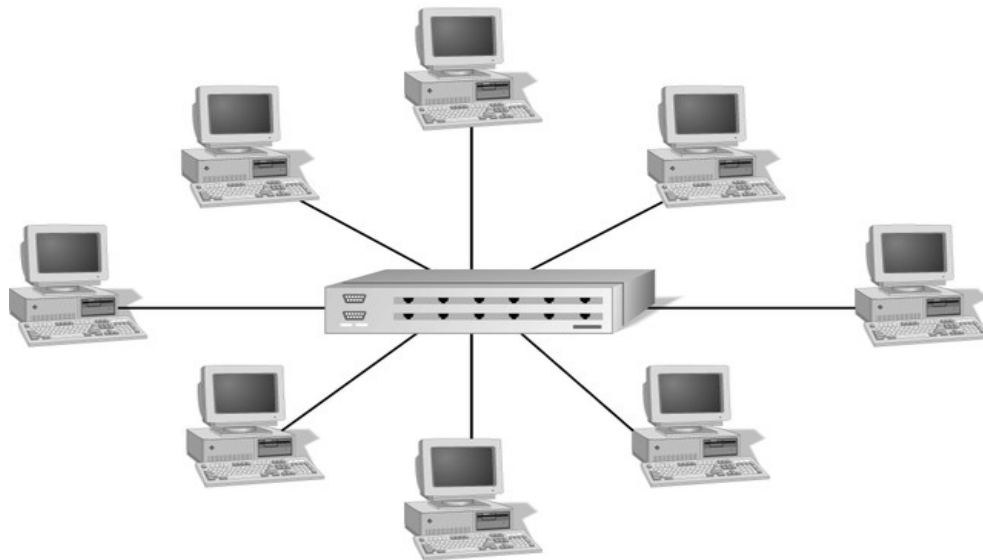
- **disadvantages**

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- signal reflection at the taps can cause degradation in quality
 - soln: limit the number and spacing of devices connected to a given length of cable
- difficult reconnection (adding new devices) and fault isolation
 - If the cable experiences a failure, the whole network goes down, which can be time-consuming and expensive to restore
 - Bus topologies are best suited for small networks because there's only so much bandwidth, and every additional node will slow transmission speeds.
 - If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.
 - Furthermore, data is “half-duplex,” which means it can't be sent in two opposite directions at the same time, so this layout is not the ideal choice for networks with huge amounts of traffic.
 - Security is very low.

Star Topology



- Every node on the network is connected through a central device called **hub or switch**.
- Is the most common Physical topology in Ethernet (LANs)



- Any single cable connects only two devices
 - Cabling problems affect two nodes at most
- Requires more cabling than ring or bus networks
 - More fault-tolerant
- Easily moved, isolated, or interconnected with other networks
 - Scalable

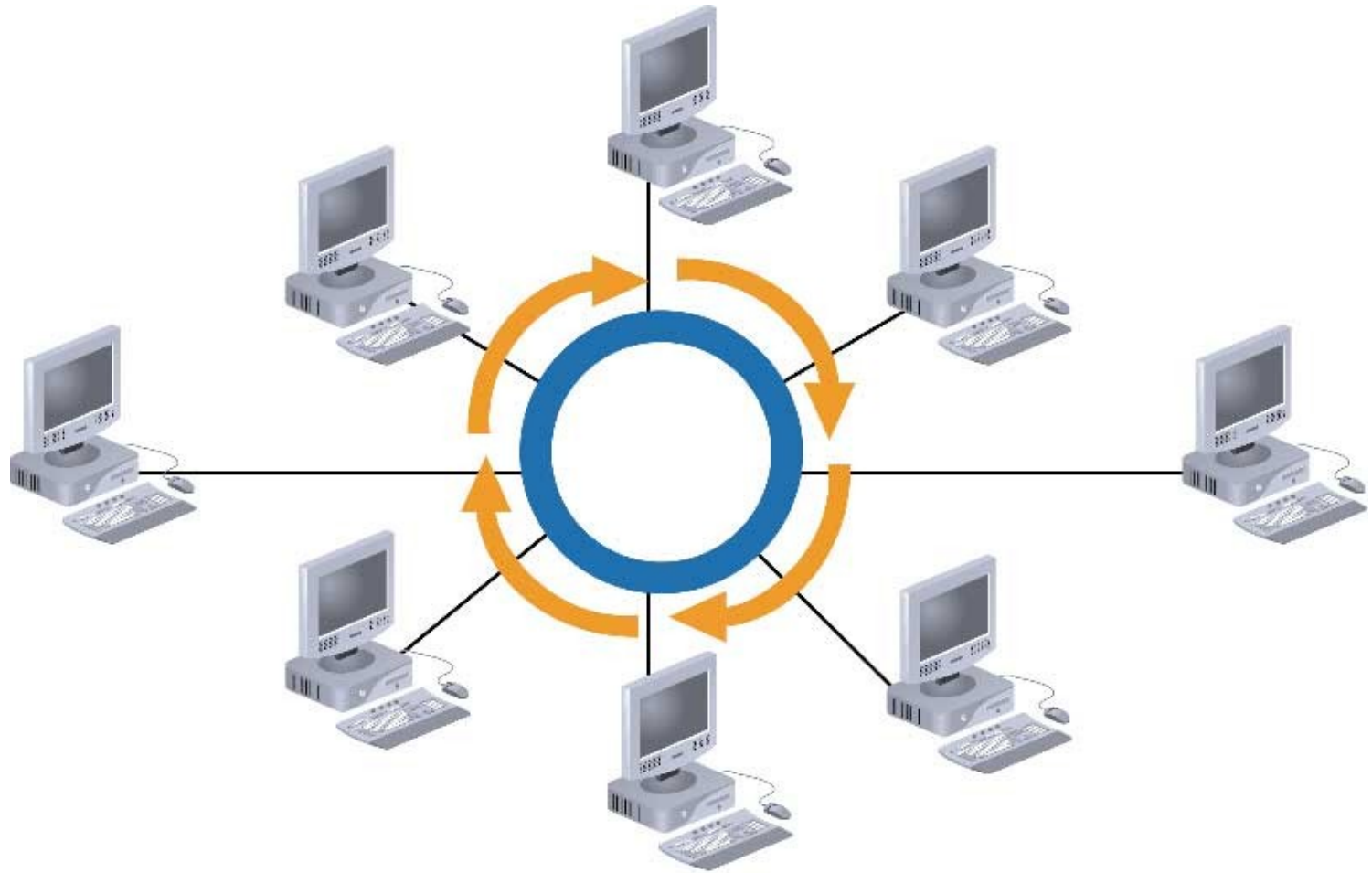
- **Advantages**

- Good option for modern networks
- Easy to manage
- Robust
- a failure of a link has no effect on others
- devices can be added, removed, and modified without taking the entire network offline. Offers opportunities for expansion
- fault identification and isolation are easy
- less expensive than mesh (but more expensive than others)

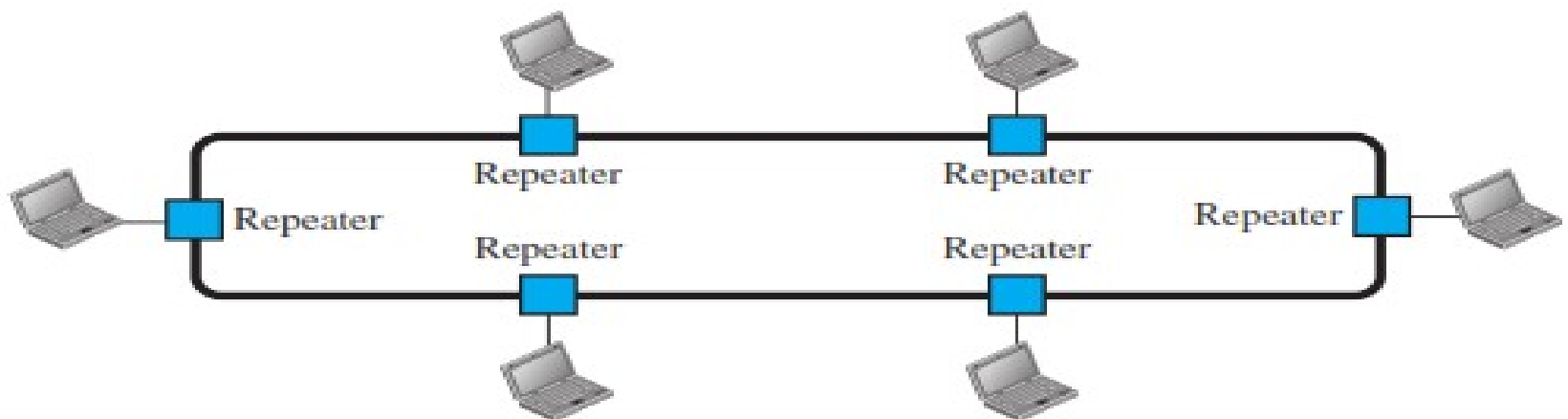
- **Disadvantages**

- Single point of failure
- Requires more cable length than a linear topology (Bus or Ring)
- More expensive than linear bus topologies because of the cost of the concentrators.

Ring topology



- each device has a dedicated point-to-point connection with only the two devices on either side of it (either physically or logically) and forms a circle.
- a signal is passed along the ring in **one direction**, from device to device, until it reaches its destination
- each device incorporates a repeater (to regenerate bits received before passing it)



■ **Advantages**

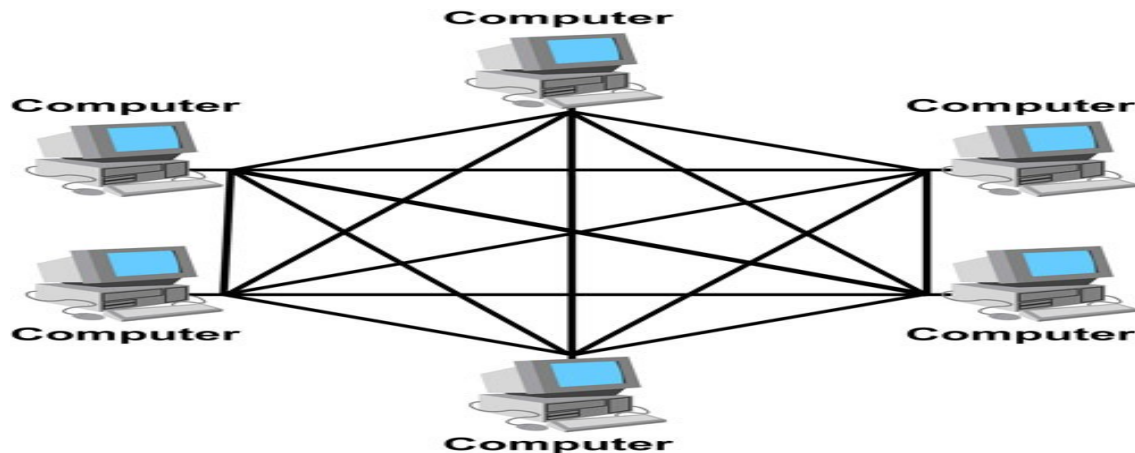
- Ring networks are moderately easy to install
- Equal access for all users
- Each workstation has full access speed to the ring
- As workstation numbers increase performance diminishes slightly
- Cable faults are easily located, making troubleshooting easier

■ **Disadvantages**

- In a simple ring, a break in the ring (such as a disabled station) can disable the entire network.
- Difficult Connections.
- Expensive.
- Requires more cable and network equipment at the start.
- Expansion to the network can cause network disruption.

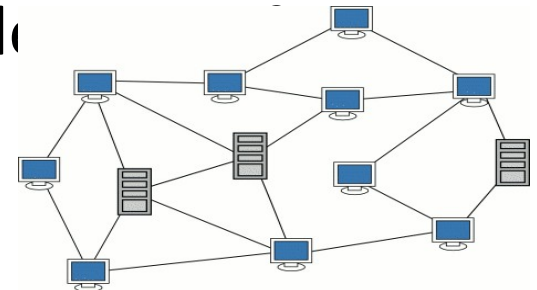
Mesh Topology

- every device has a dedicated point-to-point link to every other device
- every device must have $n-1$ I/O ports



Simple Physical Topologies

- Mesh Topology: Each computer connects to every other.
- High level of redundancy.
- every device must have $n-1$ I/O ports
- Rarely used.
 - Wiring is very complicated
 - Cabling cost is high
 - Troubleshooting a failed cable is tricky
 - A variation hybrid mesh – create point to point
- connection between specific network devices
- seen in WAN implementation.



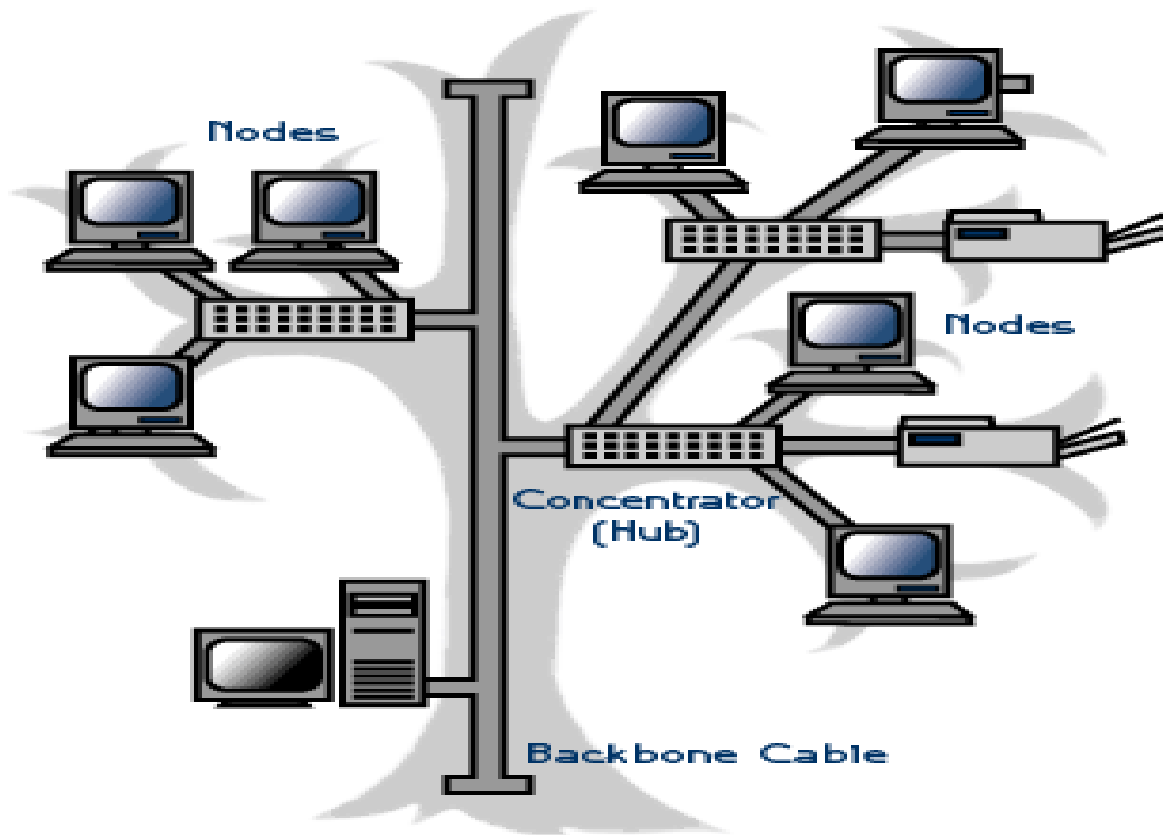
■ **Advantages**

- no traffic problem
- robust; a failure of a link has no effect on others
- There is the advantage of privacy or security
- The network can be expanded without disruption to current uses
- Point to point links make fault identification and fault isolation easy

■ **Disadvantages**

- Requires more cable than the other LAN topologies
- Complicated implementation
 - Installation and reconnection are difficult.
 - Sheer bulk of wiring can be greater than the available space can accommodate
 - Expensive

Hybrid Topologies



Hybrid Topologies

- One example of **Hybrid Topology** is **Tree topology**
- **Tree topology** is a combination of Bus and Star topology.
- It consists of groups of star-configured workstations connected to a linear bus backbone cable.
- If the backbone line breaks, the entire segment goes down
- An example of this network could be cable TV technology

Choosing a Topology

- **BUS**
 - network is small
 - network will not be frequently reconfigured
 - least expensive solution is required
 - network is not expected to grow much
- **STAR**
 - it must be easy to add/remove PCs
 - it must be easy to troubleshoot
 - network is large
 - network is expected to grow in the future
- **RING**
 - network must operate reasonably under heavy load
 - higher speed network is required
 - network will not be frequently reconfigured